Below is a complete database architecture for Auzaarbazaar as a multi-tenant B2B marketplace (Supabase Postgres + Prisma-friendly). It supports: buyers + sellers, RFQs → quotes → negotiation → orders, chat + files, verification + trust, promotions + billing, and analytics.

1) Core principles

Multi-tenant: everything belongs to an org (buyer company or seller company).

One user can belong to many orgs with roles (buyer, seller, admin).

RFQ is the main conversion object.

Listings are structured + searchable: taxonomy + attributes.

Chat and files are first-class (quotations/spec PDFs, drawings, invoices).

2) Entity map (ERD in text)

auth.users (Supabase) → profiles

orgs ↔ org_members

orgs (seller) → listings → listing_images, listing_attributes

categories → category_attributes (defines spec schema)

Buyers create rfqs → rfq_items (machine requirements)

Sellers respond with quotes → quote_items

conversations ↔ conversation_participants → messages → message_attachments

Accepted quote → orders → order_items → payments (optional)

Trust: org_verifications, reviews, response_metrics

Monetization: plans, subscriptions, promotions, promotion_events

3) Tables (recommended)

A) Identity & tenancy

profiles

id uuid pk (same as auth.users.id)

full_name text

phone text

avatar_url text

created_at timestamptz

orgs

id uuid pk

name text

org_type enum('buyer','seller','both')

gstin text null

address_id uuid null

is_verified boolean default false

created_at timestamptz

org_members

org_id uuid fk orgs

user_id uuid fk profiles

role enum('owner','admin','member','viewer')

persona enum('buyer','seller','ops') (how they use the product)

status enum('active','invited','disabled')

joined_at timestamptz PK: (org_id, user_id)

Index: (user_id), (org_id, role)

addresses

id uuid pk

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line1 text, line2 text
city text, state text, pincode text, country text default 'IN'
lat double precision null, lng double precision null
B) Taxonomy & specs system (key to search + filtering)
categories
id uuid pk
parent_id uuid fk categories null
name text
slug text unique
path_ltree ltree (optional but powerful for hierarchical queries)
category_attributes (defines spec schema per category)
id uuid pk
category_id uuid fk categories
key text (e.g. spindle_speed_rpm)
label text
type enum('number','text','boolean','select','multiselect','range')
unit text null (rpm, kW, mm)
required boolean default false
options jsonb null (for select/multiselect)
filterable boolean default true
sort_order int
C) Listings (seller supply)
listings
id uuid pk
seller_org_id uuid fk orgs
category_id uuid fk categories
title text
slug text unique
condition enum('new','used','refurbished')
price_type enum('fixed','range','contact')
price_min numeric(14,2) null
price_max numeric(14,2) null
currency text default 'INR'
location_id uuid fk addresses null
availability enum('in_stock','made_to_order','out_of_stock')
lead_time_days int null
warranty_months int null
status enum('draft','published','paused','archived')
published_at timestamptz null
created_at timestamptz
listing_attributes (EAV but controlled by category schema)
listing_id uuid fk listings
attribute_key text (matches category_attributes.key)
value jsonb (store number/text/arrays/range) PK: (listing_id, attribute_key)
GIN index: value (jsonb)
listing_images

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id uuid pk
listing_id uuid fk listings
storage_path text (Supabase Storage)
alt text null
sort_order int
created_at timestamptz
listing_documents
id uuid pk
listing_id uuid fk listings
doc_type enum('brochure','manual','spec_sheet','certificate','other')
storage_path text
created_at timestamptz
D) RFQs (buyer demand)
rfqs
id uuid pk
buyer_org_id uuid fk orgs
title text
budget_min numeric(14,2) null
budget_max numeric(14,2) null
currency text default 'INR'
delivery_location_id uuid fk addresses null
needed_by date null
status enum('draft','submitted','in_review','open','closed','cancelled')
created_at timestamptz
submitted_at timestamptz null
rfq_items
id uuid pk
rfq_id uuid fk rfqs
category_id uuid fk categories
quantity int default 1
requirements jsonb (same attribute keys as category schema; can include ranges)
notes text null
rfq_targets (who received the RFQ; supports “broadcast RFQ” + direct RFQ)
rfq_id uuid fk rfqs
seller_org_id uuid fk orgs
target_type enum('direct','broadcast','recommended')
status enum('sent','viewed','responded','declined')
last_action_at timestamptz PK: (rfq_id, seller_org_id)
E) Quotes & negotiation
quotes
id uuid pk
rfq_id uuid fk rfqs
seller_org_id uuid fk orgs
buyer_org_id uuid fk orgs
status enum('draft','sent','revised','accepted','rejected','expired','withdrawn')
subtotal numeric(14,2)

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tax numeric(14,2) null
shipping numeric(14,2) null
total numeric(14,2)
currency text default 'INR'
valid_until date null
lead_time_days int null
warranty_months int null
terms text null
created_at timestamptz
sent_at timestamptz null
quote_items
id uuid pk
quote_id uuid fk quotes
rfq_item_id uuid fk rfq_items null
listing_id uuid fk listings null (seller can propose a specific listing)
title text
quantity int
unit_price numeric(14,2)
total_price numeric(14,2)
spec_snapshot jsonb (freeze specs at quote time)
quote_attachments
id uuid pk
quote_id uuid fk quotes
storage_path text
file_name text
created_at timestamptz
negotiation_events (audit trail for pricing / terms changes)
id uuid pk
quote_id uuid fk quotes
actor_org_id uuid fk orgs
event_type enum('price_change','term_change','message','status_change')
payload jsonb
created_at timestamptz
F) Orders (optional now, but future-proof)
orders
id uuid pk
buyer_org_id uuid fk orgs
seller_org_id uuid fk orgs
rfq_id uuid fk rfqs null
quote_id uuid fk quotes null
status enum('pending','confirmed','in_progress','shipped','delivered','cancelled','disputed','completed')
total numeric(14,2)
currency text default 'INR'
created_at timestamptz
order_items
id uuid pk

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order_id uuid fk orders
listing_id uuid fk listings null
title text
quantity int
unit_price numeric(14,2)
total_price numeric(14,2)
spec_snapshot jsonb
payments (if you add escrow/online payment)
id uuid pk
order_id uuid fk orders
provider text (razorpay/stripe/etc.)
status enum('pending','paid','failed','refunded')
amount numeric(14,2)
currency text
provider_ref text null
created_at timestamptz
G) Messaging (chat that ties to RFQs/quotes)
conversations
id uuid pk
kind enum('rfq','quote','order','general')
rfq_id uuid null
quote_id uuid null
order_id uuid null
created_at timestamptz
conversation_participants
conversation_id uuid fk conversations
org_id uuid fk orgs
joined_at timestamptz PK: (conversation_id, org_id)
messages
id uuid pk
conversation_id uuid fk conversations
sender_user_id uuid fk profiles
sender_org_id uuid fk orgs
body text
created_at timestamptz
read_at timestamptz null
message_attachments
id uuid pk
message_id uuid fk messages
storage_path text
file_name text
H) Trust & verification
org_verifications
id uuid pk
org_id uuid fk orgs
type enum('gst','address','phone','bank','other')

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status enum('pending','verified','rejected')
evidence_path text null
created_at timestamptz
reviews
id uuid pk
order_id uuid fk orders null
buyer_org_id uuid fk orgs
seller_org_id uuid fk orgs
rating int check (rating between 1 and 5)
comment text null
created_at timestamptz
seller_metrics_daily (denormalized for speed)
seller_org_id uuid
date date
rfqs_received int
quotes_sent int
response_time_avg_seconds int
orders_closed int
revenue numeric(14,2) PK: (seller_org_id, date)
I) Monetization (plans, subscriptions, promotions)
plans
id uuid pk
code text unique (free/pro/growth/enterprise)
features jsonb
price_monthly numeric(14,2)
currency text default 'INR'
subscriptions
id uuid pk
org_id uuid fk orgs
plan_id uuid fk plans
status enum('active','past_due','cancelled')
started_at timestamptz
ends_at timestamptz null
provider text null
provider_ref text null
promotions
id uuid pk
seller_org_id uuid fk orgs
listing_id uuid fk listings
type enum('featured','top_search','category_spotlight')
status enum('scheduled','active','ended','cancelled')
starts_at timestamptz
ends_at timestamptz
meta jsonb (target category/region)
4) Indexing & performance essentials
listings: index (category_id, status), (seller_org_id), (published_at desc)

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listing_attributes: GIN on value, btree on attribute_key
rfqs: index (buyer_org_id, status, created_at desc)
rfq_targets: index (seller_org_id, status, last_action_at desc)
quotes: index (rfq_id), (seller_org_id, status), (buyer_org_id, status)
messages: index (conversation_id, created_at desc)
Optional: categories.path_ltree GIST index if using ltree
5) Supabase RLS (high-level rules)
You'll implement RLS on every multi-tenant table using "org membership".
Pattern:
A user can SELECT/INSERT/UPDATE rows where:
org_id or buyer_org_id or seller_org_id is in the user's org memberships.
Minimal helpers:
org_members is the source of truth.
Create a SQL function: is_org_member(org_id uuid) returns boolean.
Examples:
Seller can edit listings only if seller_org_id is one of their orgs.
Buyer can view their RFQs only if buyer_org_id is one of their orgs.
Seller can view RFQs only if they are in rfq_targets for that RFQ.
6) Storage architecture (ties into DB)
Store files in Supabase Storage, but DB stores paths.
Buckets:
listings/ (images + docs)
rfqs/ (drawings/specs)
quotes/ (quotation PDFs)
orgs/ (verification docs)
messages/ (chat attachments)
DB fields: storage_path text everywhere, never raw URLs.
7) Prisma modeling notes (so it's easy to code)
Use @db.Uuid, @db.Timestamptz(6), Json.
Many-to-many: org_members, conversation_participants.
Keep enums in Prisma for status fields to prevent invalid states.
8) What to build first (MVP order)
orgs, org_members, profiles
categories, category_attributes
listings, listing_attributes, listing_images
rfqs, rfq_items, rfq_targets
quotes, quote_items, basic chat (conversations/messages)
verification + reviews + analytics tables
billing + promotions

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